# Active Record Query Interface

This guide covers different ways to retrieve data from the database using Active Record.

If you're used to using raw SQL to find database records, then you will generally find that there are better ways to carry out the same operations in Rails. Active Record insulates you from the need to use SQL in most cases.

Code examples throughout this guide will refer to one or more of the following models:

All of the following models use `id` as the primary key, unless specified otherwise.

```
class Client < ActiveRecord::Base
  has_one :address
  has_many :orders
  has_and_belongs_to_many :roles
end
class Address < ActiveRecord::Base
  belongs_to :client
end
class Order < ActiveRecord::Base
  belongs_to :client, counter_cache: true
end
class Role < ActiveRecord::Base
  has_and_belongs_to_many :clients
end
```

Active Record will perform queries on the database for you and is compatible with most database systems (MySQL, PostgreSQL and SQLite to name a few). Regardless of which database system you're using, the Active Record method format will always be the same.

## 1 Retrieving Objects from the Database

To retrieve objects from the database, Active Record provides several finder methods. Each finder method allows you to pass arguments into it to perform certain queries on your database without writing raw SQL.

The methods are:

- `bind`
- `create_with`
- `distinct`
- `eager_load`
- `extending`
- `from`
- `group`
- `having`
- `includes`
- `joins`
- `limit`
- `lock`